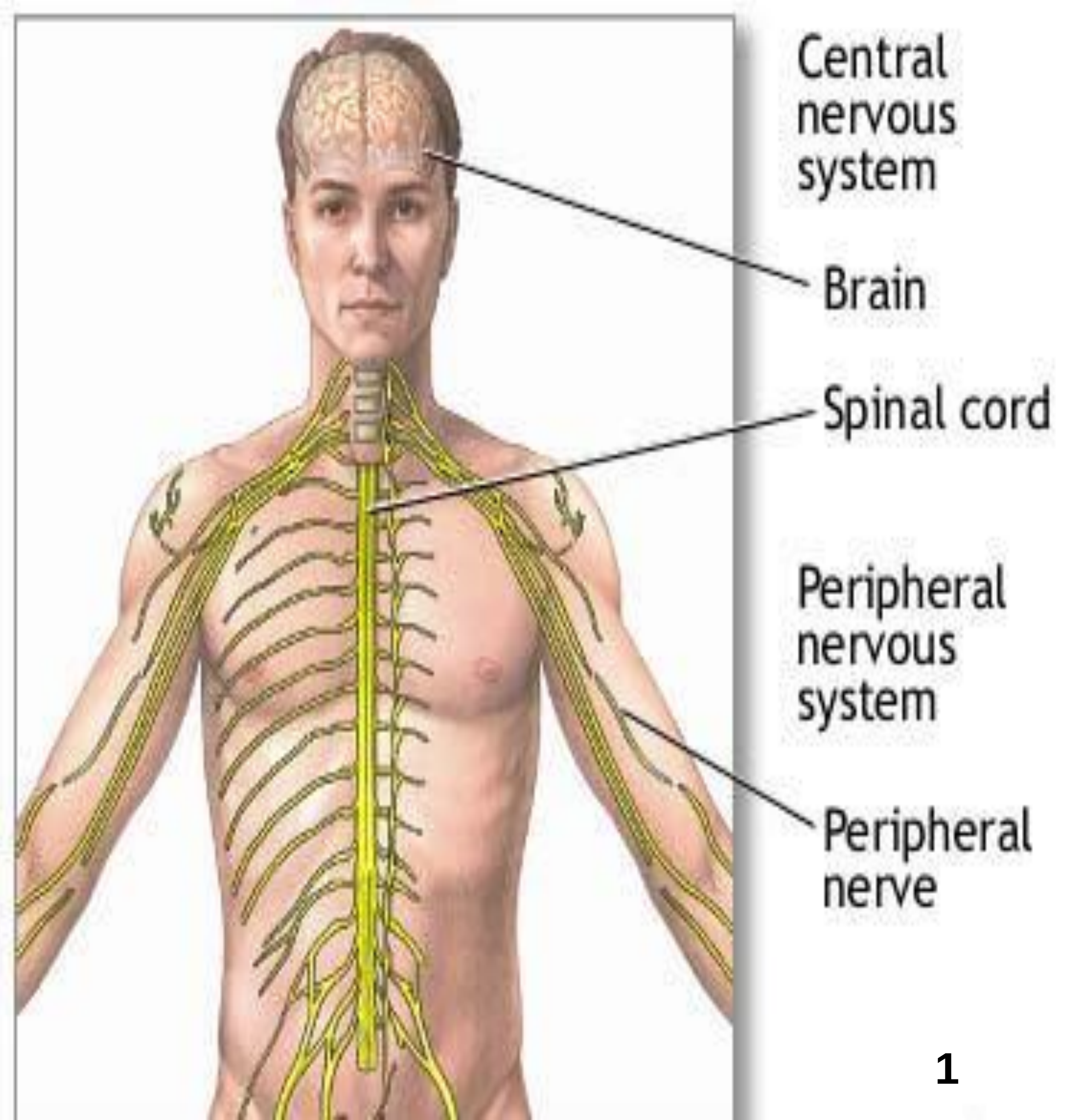
The Nervous System:

The Nervous System is divided into:

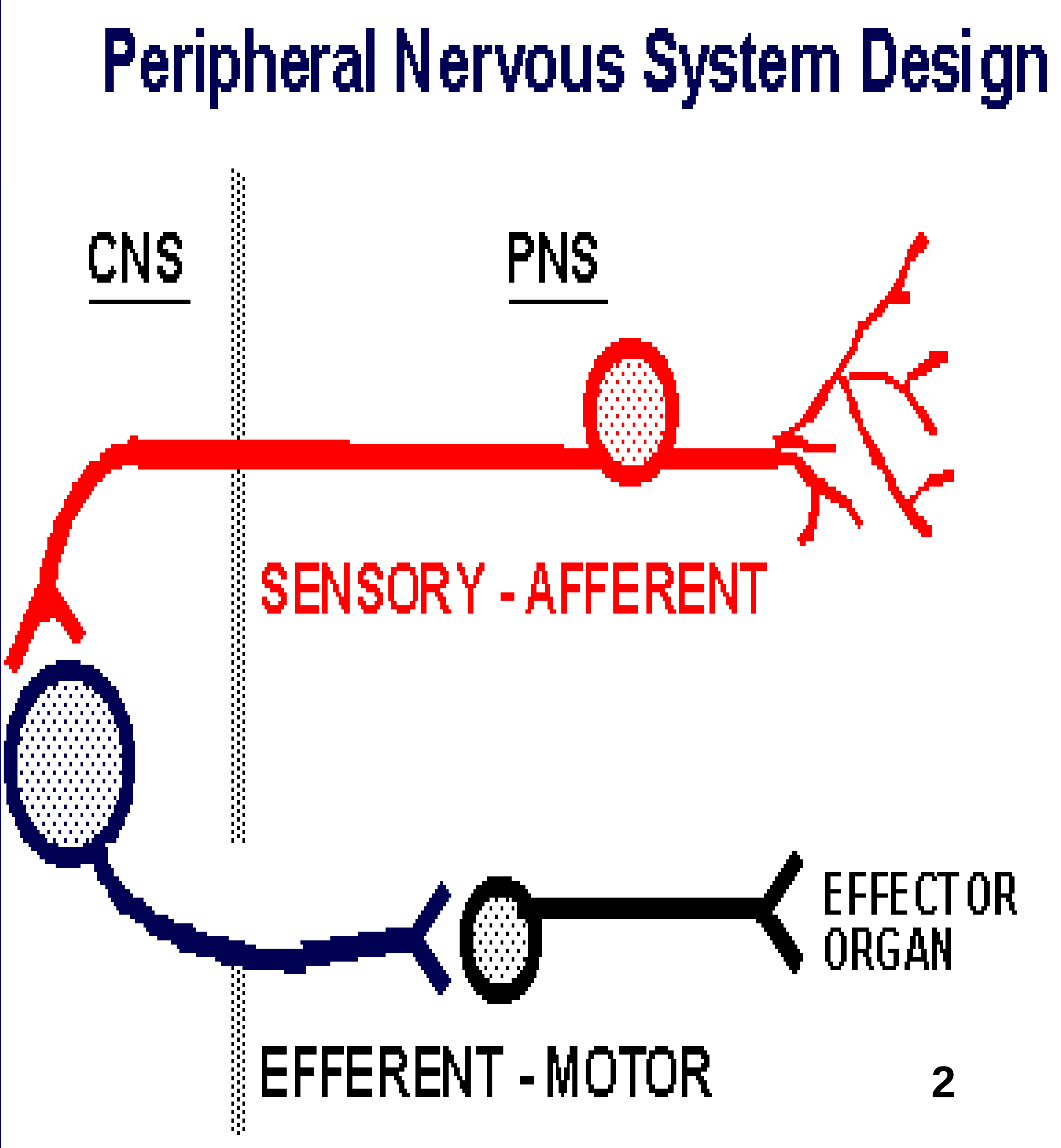
- Central nervous system (CNS): consists of brain and spinal cord.
- Peripheral nervous system (PNS) consists of afferent and efferent nerves.



The peripheral nervous system:

The PNS consists of :

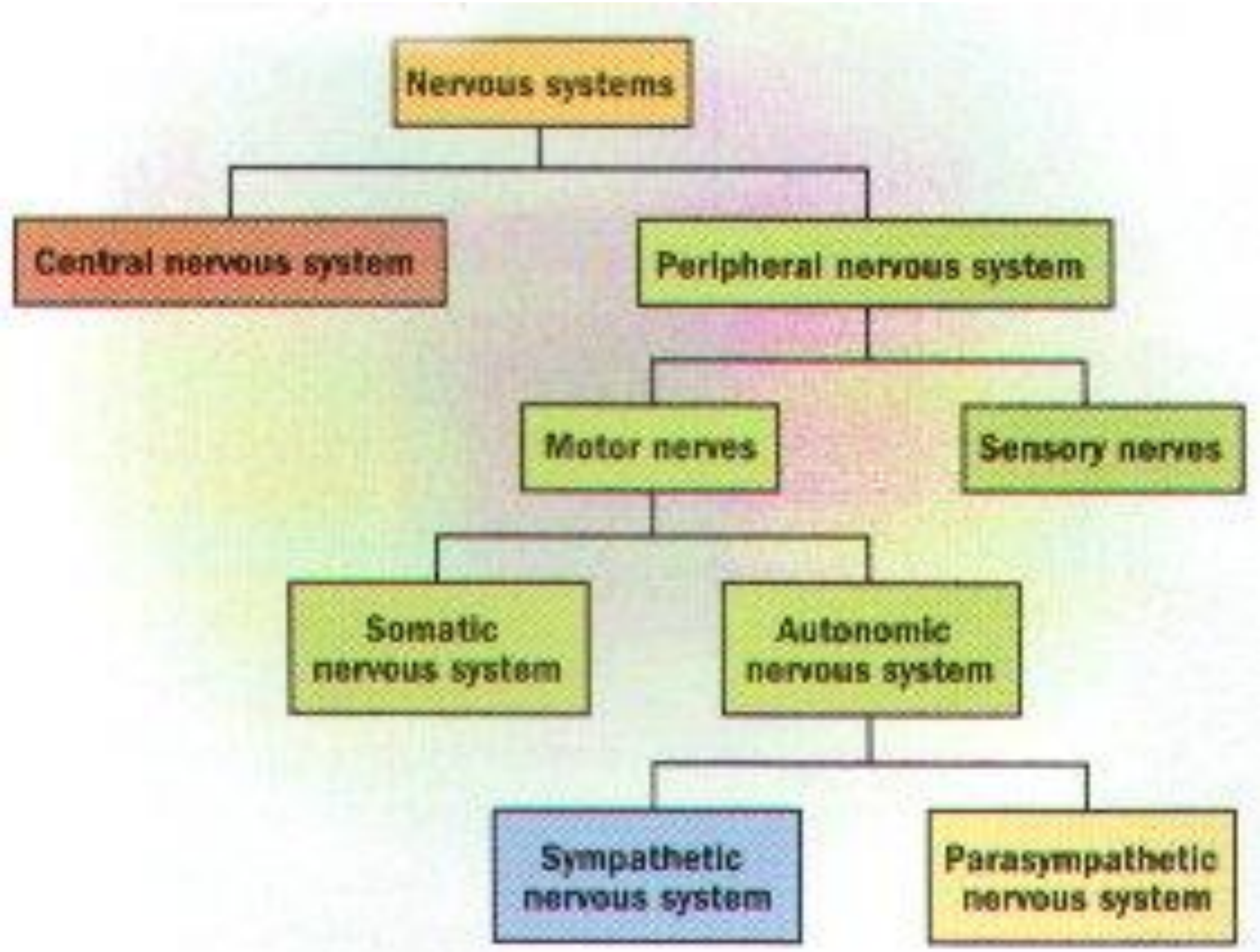
- Sensory(Afferent) neurons: running from receptors that inform CNS of stimuli.
- Motor (Efferent) neurons: running from CNS to muscles and glands (effector organs).



The Peripheral Nervous system:

- 31 pairs of spinal nerves (all are mixed sensory & motor).

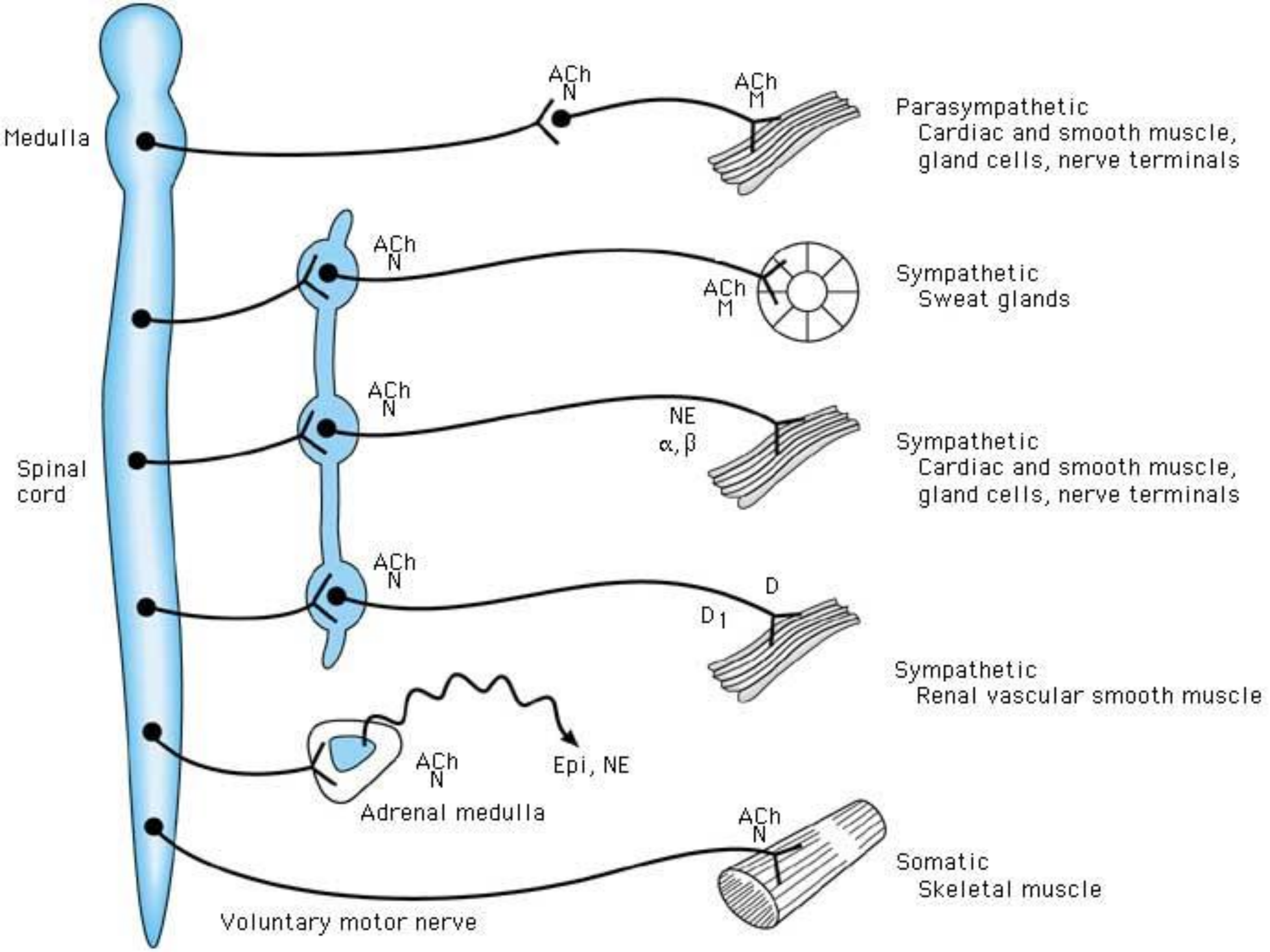
- 12 pairs of cranial nerves (some are pure sensory, some are pure motor , but most are mixed) .


Nervous systems
Central nervous system
Peripheral nervous system
Motor nerves
Sensory nerves
Somatic nervous system
Autonomic nervous system
Sympathetic nervous system
Parasympathetic nervous system

Efferent (Motor) Nervous System:

The motor part of the PNS is divided into:

- Somatic motor nervous system .
- Autonomic Nervous System (ANS).

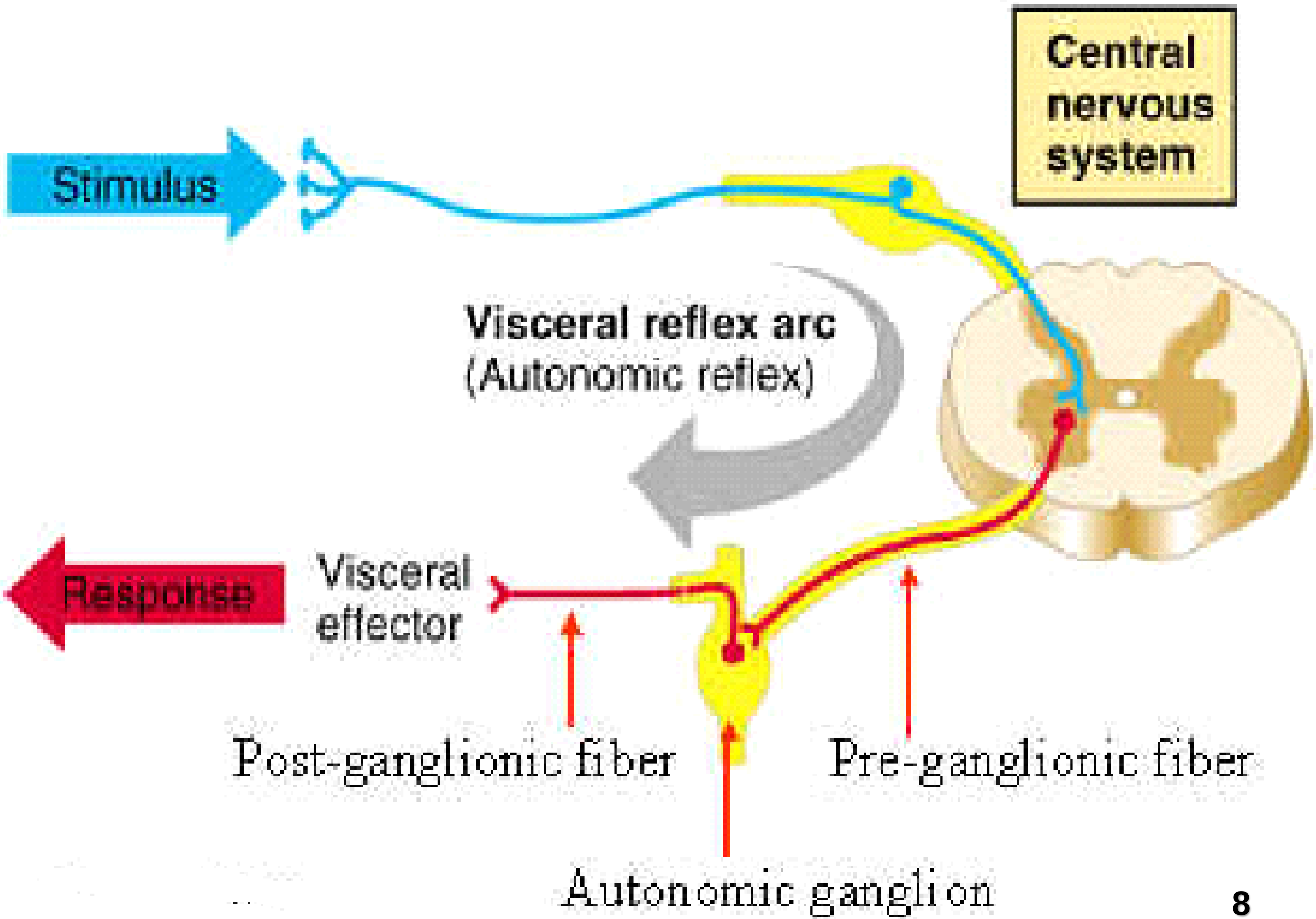

Medulla
ACh
N
ACh
M
Parasympathetic
Cardiac and smooth muscle, gland cells, nerve terminals
ACh
N
ACh
M
Sympathetic
Sweat glands
Spinal cord
ACh
N
NE
α, β
Sympathetic
Cardiac and smooth muscle, gland cells, nerve terminals
ACh
N
D
D1
Sympathetic
Renal vascular smooth muscle
ACh
N
Epi, NE
Adrenal medulla
ACh
N
Somatic
Skeletal muscle
Voluntary motor nerve

Autonomic Nervous System

The Autonomic Nervous System (ANS) controls many of the involuntary activities of the body, for example digestion, sweating ,,,,,

The ANS acts on :-

- Cardiac muscle of the heart,
- Smooth muscle of the gut,
- Blood vessels,
- Urogenital tract
- Secretory glands


Central nervous system
Stimulus
Visceral reflex arc
(Autonomic reflex)
Response
Visceral effector
Post-ganglionic fiber
Pre-ganglionic fiber
Autonomic ganglion

Autonomic Nervous System Divisions:

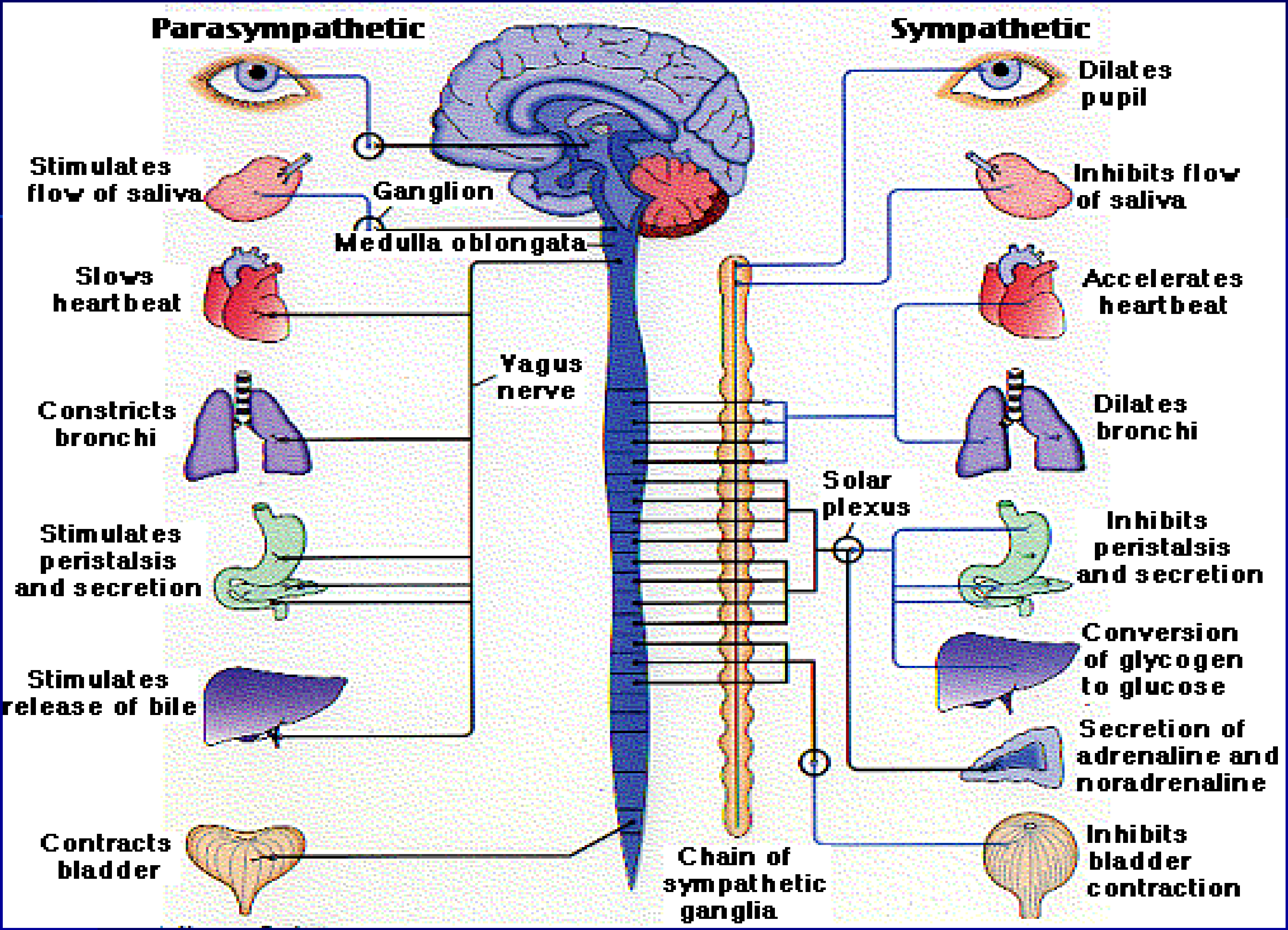
- The ANS is divided into two parts:-

  -The Sympathetic (Thoraco – Lumbar) Nervous System

  and

  -The Parasympathetic (Cranio – Sacral) Nervous System.

  These usually have opposite (antagonistic) effects on the organs which they supply.

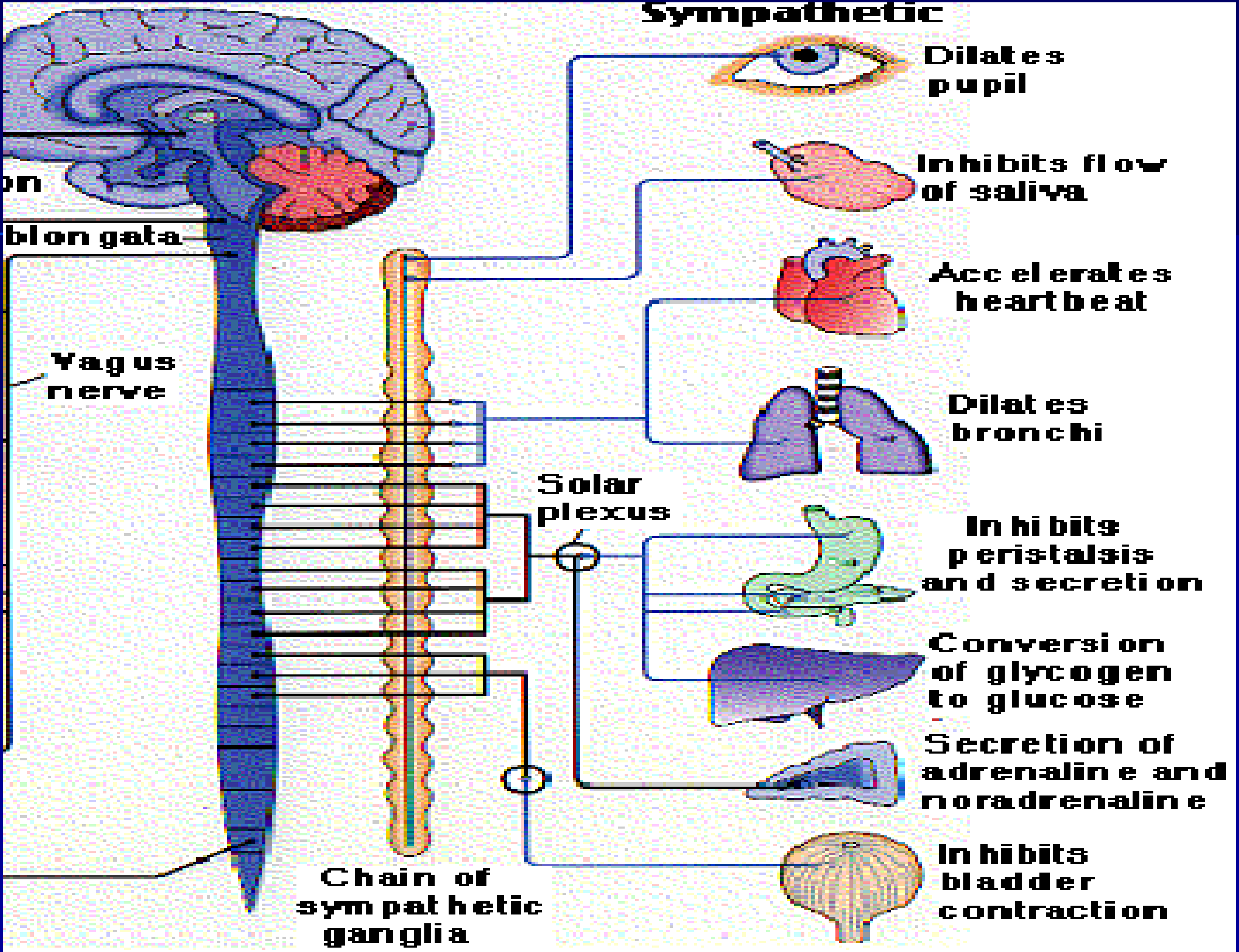

Parasympathetic
Sympathetic
Stimulates flow of saliva
Ganglion
Medulla oblongata
Slows heartbeat
Vagus nerve
Constricts bronchi
Stimulates peristalsis and secretion
Stimulates release of bile
Contracts bladder
Dilates pupil
Inhibits flow of saliva
Accelerates heartbeat
Dilates bronchi
Solar plexus
Inhibits peristalsis and secretion
Conversion of glycogen to glucose
Secretion of adrenaline and noradrenaline
Inhibits bladder contraction
Chain of sympathetic ganglia

The Sympathetic
"Fight or Flight"

- Dominates in emergency situations:-

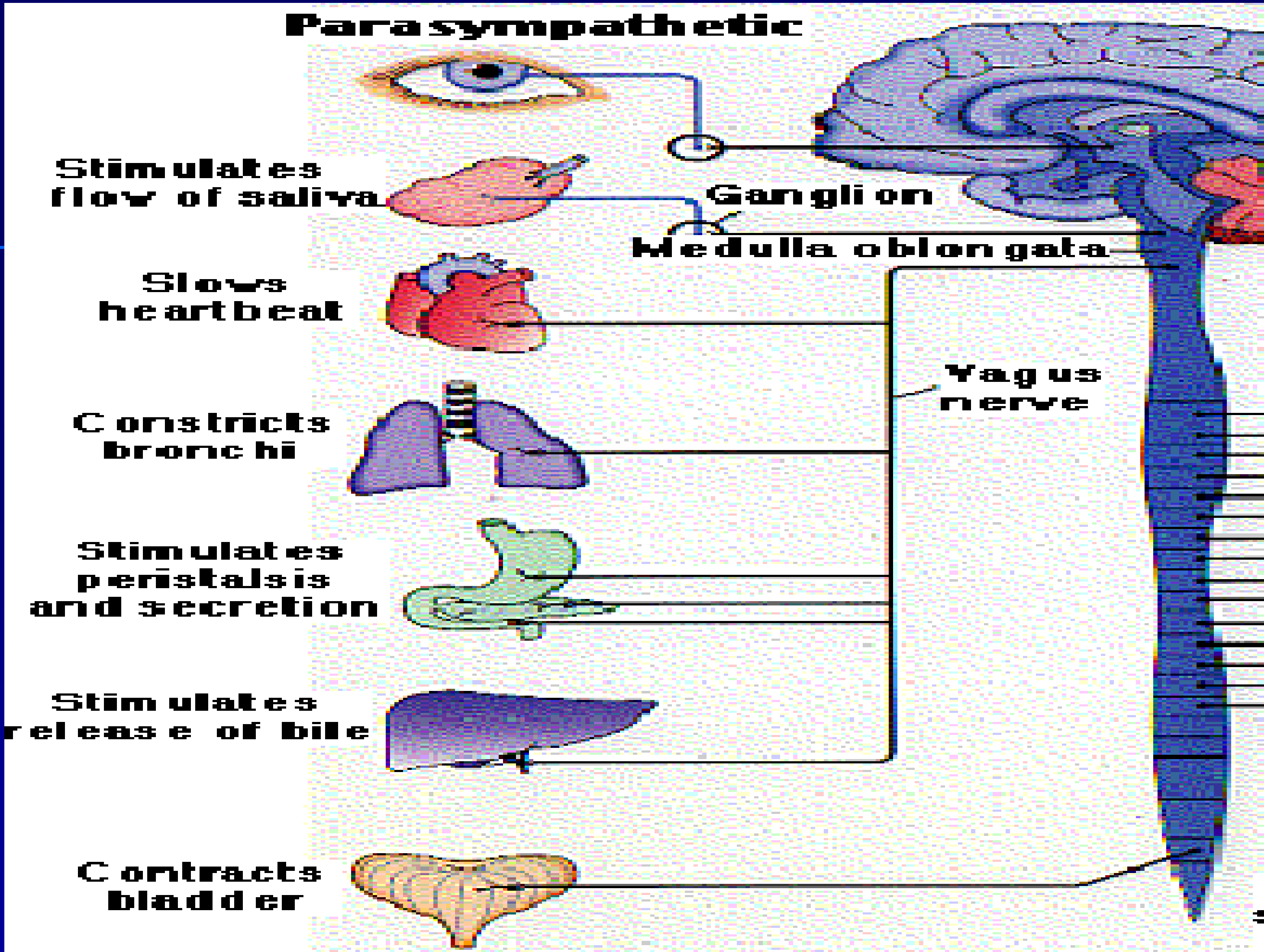
The Whole sympathetic system tends to "go off" together Causing:-

- Redistribution of the blood.
- Increased oxygen and energy supply to muscles .
- Increased sweating.
- Slowing of non essential involuntary functions.


Sympathetic
Dilates pupil
Inhibits flow of saliva
Accelerates heartbeat
Dilates bronchi
Inhibits peristalsis and secretion
Conversion of glycogen to glucose
Secretion of adrenaline and noradrenaline
Inhibits bladder contraction
on
blongata
Vagus nerve
Solar plexus
Chain of sympathetic ganglia

The Parasympathetic N.S "Rest and Digest"

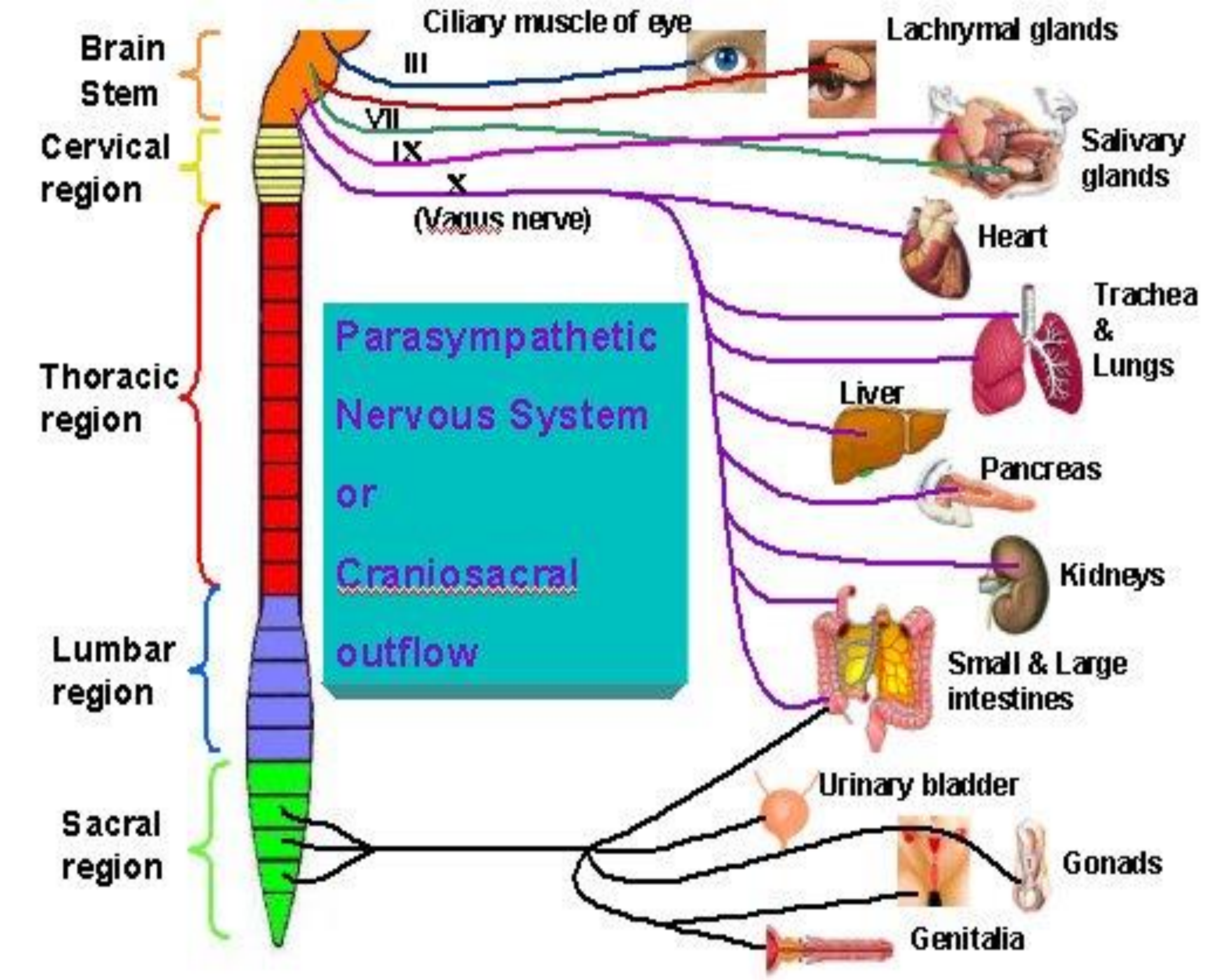
- Normal maintenance of the body-
- It promotes secretions and mobility of different parts of the digestive tract.
- Also involved in urination, defecation.
- Does not "go off" together; activities initiated when appropriate .


Parasympathetic
Stimulates flow of saliva
Ganglion
Medulla oblongata
Slows heartbeat
Vagus nerve
Constricts bronchi
Stimulates peristalsis and secretion
Stimulates release of bile
Contracts bladder

Parasympathetic:

- The vagus nerve (cranial number 10) is the chief parasympathetic nerve .

- Other cranial parasympathetic nerves are: III (oculomotor), VII (facial) and IX (glossopharyngeal)



Ciliary muscle of eye
Lachrymal glands
Brain Stem
III
VII
IX
X
(Vagus nerve)
Cervical region
Salivary glands
Heart
Thoracic region
Trachea & Lungs
Parasympathetic Nervous System or Craniosacral outflow
Liver
Pancreas
Kidneys
Lumbar region
Small & Large intestines
Urinary bladder
Sacral region
Gonads
Genitalia

Chemical Transmitters:

	Sympathetic	PARAS
Preganglionic fibers:	A.Ch.	A.Ch.
Post-ganglionic fibers:	Noradr.	A.Ch.
Skeletal fibers:	A.Ch.	

Actions of Autonomic Nervous System

Organ	Sympathetic Stimulation	Parasympathetic Stimulation
Heart	Increased heart rate beta$_1$ (& beta$_2$)	Decreased heart rate
	Increased force of contraction beta$_1$ (& beta$_2$)	Decreased force of contraction
	Increased conduction velocity	Decreased conduction velocity
Arteries	Constriction (alpha$_1$)	Dilation
	Dilation (beta$_2$)	
Veins	Constriction (alpha$_1$)	
	Dilation (beta$_2$)	
Lungs	Bronchial muscle relaxation (beta$_2$)	Bronchial muscle contraction
		Increased bronchial gland secretions
Gastro-intestinal tract	Decreased motility (beta$_2$)	Increased motility
	Contraction of sphincters (alpha)	Relaxation of sphincters

<table>
<tr><td rowspan="3">Liver</td><td>Glycogenolysis (beta$_2$ & alpha)</td><td rowspan="3">Glycogen synthesis</td></tr>
<tr><td>Gluconeogenesis (beta$_2$ & alpha)</td></tr>
<tr><td>Lipolysis (beta$_2$ & alpha)</td></tr>
<tr><td>Kidney</td><td>Renin secretion (beta$_2$)</td><td></td></tr>
<tr><td rowspan="2">Bladder</td><td>Detrusor relaxation (beta$_2$)</td><td>Detrusor contraction</td></tr>
<tr><td>Contraction of sphincter (alpha)</td><td>Relaxation of sphincter</td></tr>
<tr><td rowspan="2">Uterus</td><td>Contraction of pregnant uterus (alpha)</td><td rowspan="2"></td></tr>
<tr><td>Relaxation of pregnant and non-pregnant uterus (beta$_2$)</td></tr>
<tr><td rowspan="2">Eye</td><td rowspan="2">Dilates pupil (alpha)</td><td>Constricts pupil</td></tr>
<tr><td>Increased lacrimal gland secretions</td></tr>
<tr><td>Submandibular & parotid glands</td><td>Viscous salivary secretions (alpha)</td><td>Watery salivary secretions</td></tr>
</table>

Control of Autonomic N.S.:

- The responses of the ANS are coordinated by two subconscious brain regions:

  - The medulla oblongata.

  and

  - The hypothalamus.

- A certain amount of conscious control can be exerted over the ANS and the organs under its control e.g. heart and gut .